Humanitarian Logistics Hub Network Design Under Uncertainty: A Multi-Objective Model and Priority-Based Genetic Algorithm for Central Coastal Vietnam

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Abstract. Conventional disaster relief models often assume uninterrupted road networks, making them inadequate for Central Vietnam, where frequent floods severely disrupt infrastructure. We formulate the Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP), a two-stage stochastic program that jointly minimizes expected logistics cost and expected maximum deprivation cost over a set of flood scenarios. The model explicitly accounts for broken inter-hub links, multi-modal transport (land, water, and air), pre-positioned inventory, as well as adaptive lateral transshipments and reactive hub activations to recover from severe localized deficits. To solve MO-IHLNDP, we propose PB-NSGA (Priority-Based Nondominated Sorting Genetic Algorithm), which confines the chromosome to first-stage hub decisions and reconstructs all scenario-dependent second-stage variables through a shared priority-based decoder, reducing search-space complexity from $O(|\mathcal{S}| \times |\mathcal{V}|)$ to $O(|\mathcal{H}| + |\mathcal{I}|)$. Experiments on standard benchmarks and a realistic CV-Small instance validate near-optimal accuracy against exact Pareto fronts. On the full-scale CV-Large case study under matched 20-seed run budgets, PB-NSGA uniquely achieves 100% out-of-sample feasibility on adversarial extreme scenarios and delivers a quantified cost-deprivation trade-off and a stable hub investment hierarchy for four flood-prone provinces in Central Vietnam.

Keywords: Humanitarian logistics · Hub location · Multi-objective optimization · Stochastic programming · Disaster relief · Evolutionary algorithm

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1 Introduction

1.1 Problem

In 2025, a relentless sequence of typhoons and record-breaking floods in Central Vietnam resulted in massive fatalities and an estimated 98.7 trillion VND in economic losses [1].

Humanitarian operations during this period faced significant challenges. Most critically, national logistics infrastructure suffered severe disruption: the North-South Railway was paralyzed by track erosion and submersion, forcing the cancellation of 170 trains, while key arteries such as Vietnam National Route 1 were severed by landslides, isolating communities for days [1]. These events demonstrate that during a major catastrophe, the assumption of a fully connected transport network is fundamentally flawed – the physical network becomes "incomplete", characterized by broken links and isolated nodes. Compounding this, resource scarcity arising from poor planning and uncertain supply and demand further strains relief efforts. Disaster Relief Network Design (DRND) bridges the preparedness and response phases by optimizing facility locations and relief routing [2].

1.2 Related Works

Humanitarian logistics prioritises speed and equity over cost minimisation [3]. Under uncertainty, stochastic programming and robust optimisation are the dominant paradigms [4,5], with deprivation cost (an exponential penalty on victim waiting time) capturing social welfare [6]. Hub-and-spoke structures consolidate relief flows efficiently [7]; for disaster settings, *incomplete* topologies in which infrastructure damage severs inter-hub arcs are more realistic [8]. Sangsawang and Chanta [9] established the flood-specific hub-location baseline (single-objective, deterministic, Thailand), and multi-modal extensions further address heterogeneous transport modes [10].

Two-stage stochastic programming separates pre-disaster design from post-disaster recourse [11]; robust formulations and prepositioning models outperform deterministic baselines under demand and supply uncertainty [12, 13]. NSGA-II [14] is the standard multi-objective framework in humanitarian logistics, valued for parameter-free Pareto ranking and elitist selection [10, 15].

Research gap and contributions. To the best of our knowledge, no existing model simultaneously addresses all four of the following: (1) incomplete hub networks with severed inter-hub arcs, (2) two-stage stochastic programming under flood uncertainty, (3) bi-objective optimisation of logistics cost *and* deprivation cost, and (4) a calibrated flood-disaster case study for Central Vietnam. Deterministic models [9] are the closest precedent: they assume full network connectivity and optimise a single cost objective, causing them to fail structurally when floods sever road links, the very scenario we target. Stochastic models [11, 12] capture demand and supply uncertainty but do not model link failures in the hub sub-network, and omit the deprivation cost that captures social equity across victim groups. This paper closes all four gaps simultaneously by formulating MO-IHLNDP and solving it via PB-NSGA, a priority-based [16] evolutionary algorithm whose encoder-decoder design efficiently handles the coupled first- and second-stage decisions of the two-stage stochastic program.

2 Problem Description and Mathematical Model

We model the network design problem as a capacitated, single-allocation MO-IHLNDP. The network comprises **origins** (supply sources), **demands** (victim groups), and **hubs** of two types: planned (pre-disaster) and reactive (post-disaster, dynamically opened).

From a practical standpoint, DRND involves two critical uncertainty factors. First, infrastructure disruption – damaged roads and buildings – can disable pre-programmed plans, necessitating a dynamic solution [12]. Second, the exact locations of demands, the total number of victims, and the availability of origins are all highly uncertain during rescue operations [11]. To account for these, we employ a two-stage stochastic programming approach across multiple scenarios, each assigning a specific risk factor per area, and assume that Search and Rescue operations rely heavily on dynamic external supply sources, as pre-positioned hubs may not fully satisfy uncertain demands under extreme scenarios.

The model integrates the deprivation cost [6] for social equity, and Daganzo's continuous approximation (CA) [17] to tractably estimate last-mile rescue times.

2.1 Notation

Sets: $\mathcal{H}$, $\mathcal{I}$, $\mathcal{J}$ index candidate hubs ($k, h \in \mathcal{H}$), demand nodes ($i \in \mathcal{I}$), and relief origins ($j \in \mathcal{J}$); $\mathcal{S}$ and $\mathcal{M}$ index disaster scenarios and transport modes; $\mathcal{V} = \mathcal{H} \cup \mathcal{I} \cup \mathcal{J}$ is the full node set.

First-phase parameters (pre-disaster, strategic): fixed hub establishment cost F_k; unit holding cost c_k and inventory capacity κ_k; inter-hub flow discount α; risk acceptability threshold χ; arc transport cost and travel time C_{uvm}, τ_{uvm}; local routing cost C_m, vehicle capacity Q_m per mode, and relief-items-per-person conversion ratio γ.

Second-phase parameters (post-disaster, scenario s with probability π_s): reactive hub activation cost F^a_{ks}; hub round-trip processing time τ_{ks}; arc accessibility $a_{uvms} \in \{0, 1\}$ (1 if traversable under scenario s); node risk index r_{us}; planned-hub availability $v_{ks} \in \{0, 1\}$ with $v_{ks} = 1 \iff r_{ks} \leq \chi$, indicating whether a planned hub at k is operable under scenario s; deprivation sensitivity $\lambda_{is} = \lambda_0(1 + r_{is})$, up-weighting vulnerable populations; demand D_{is} and supply O_{js}; pre-computed cheapest transport cost $c_{jks} = \min_{m|a_{jkms}=1} C_{jkm}$ and Daganzo CA last-mile routing cost Θ_{kis} (circuity Φ, group size η, grid area A_i).

Decision variables: $x_k \in \{0, 1\}$ and $y_{ks} \in \{0, 1\}$ activate planned and reactive hubs; $z_{iks}, z_{jks} \in \{0, 1\}$ assign demand nodes and origins to hubs per scenario;

$q_k \geq 0$ is pre-positioned inventory quantity; $f_{khms} \geq 0$ is inter-hub lateral transshipment flow.

2.2 Objectives

Objective 1: Minimize Total Expected Logistics Cost (Z_1)

$$\min \quad Z_1 = \sum_k F_k x_k + \sum_k c_k q_k + \sum_s \pi_s \left[\sum_k F^a_{ks} y_{ks} + \sum_{j,k} c_{jks} O_{js} z_{jks} + \sum_{k,h,m} \alpha C_{khm} f_{khms} + \sum_{k,i} \Theta_{kis} z_{iks} \right] \tag{1}$$

The first-stage terms are pre-disaster strategic costs (hub establishment and inventory holding); the scenario-weighted second-stage terms cover, in order, reactive hub setup, origin-to-hub supply flow, discounted inter-hub lateral transshipment, and grid-based last-mile CA routing.

where Θ_{kis} is the pre-computed Daganzo CA routing cost [17] (line-haul plus local detour):

$$\Theta_{kis} = \min_{m \in \mathcal{M} | a_{ikms}=1} \left(2 \cdot C_{kim} \cdot \lceil D_{is}/Q_m \rceil + C_m \cdot \Phi \sqrt{\lceil D_{is}/\eta \rceil \cdot A_i} \right)$$

Objective 2: Minimize Expected Maximum Deprivation Cost (Z_2)

$$\min \quad Z_2 = \sum_{s \in \mathcal{S}} \pi_s \left(\max_{i \in \mathcal{I}} \left[D_{is} \cdot \left(e^{\lambda_{is} \cdot \Omega_{is}} - 1 \right) \right] \right) \tag{2}$$

$$\Omega_{is} = \sum_{k \in \mathcal{H}} z_{iks} \left(\tau_{ks} + 2 \cdot \min_{m \in \mathcal{M} | a_{ikms}=1} \tau_{kim} \right)$$

To enforce equity, Equation (2) applies an exponential penalty to waiting time Ω_{is}, scaled by vulnerability $\lambda_{is} = \lambda_0(1 + r_{is})$. Because Constraint (7) enforces single-allocation, the non-linear deprivation cost reduces to a static parameter C^{dep}_{iks} for each valid (i,k) pair, allowing exact linearization via the auxiliary bounding variable $W_s \geq C^{\text{dep}}_{iks} \cdot z_{iks}$:

$$C^{\text{dep}}_{iks} = D_{is} \cdot \left(e^{\lambda_{is} \cdot \left(\tau_{ks} + 2 \cdot \min_{m \in \mathcal{M} | a_{ikms}=1} \tau_{kim} \right)} - 1 \right) \tag{3}$$

The auxiliary variable $W_s \geq C^{\text{dep}}_{iks} \cdot z_{iks}$ linearizes the max; the complete model minimizes Z_1, Z_2 subject to:

2.3 Constraints

$$x_k + y_{ks} \leq 1 \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{4}$$
$$q_k \leq \kappa_k \cdot x_k \quad \forall k \in \mathcal{H} \tag{5}$$

$$r_{ks} \cdot y_{ks} \leq \chi \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{6}$$

$$\sum_{k\in\mathcal{H}} z_{iks} = 1,\ \sum_{k\in\mathcal{H}} z_{jks} = 1 \quad \forall i\in\mathcal{I},\ j\in\mathcal{J},\ s\in\mathcal{S} \tag{7}$$

$$z_{iks},\ z_{jks} \leq v_{ks} \cdot x_k + y_{ks} \quad \forall i, j, k, s \tag{8}$$

$$z_{iks} \leq \textstyle\sum_m a_{ikms},\ z_{jks} \leq \sum_m a_{jkms} \quad \forall i, j, k, s \tag{9}$$

$$f_{khms} \leq M \cdot a_{khms} \quad \forall k, h \in \mathcal{H},\ m \in \mathcal{M},\ s \in \mathcal{S} \tag{10}$$

$$\begin{aligned}\sum_{i\in\mathcal{I}} \gamma D_{is} z_{iks} + \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} f_{khms} \leq q_k &+ \sum_{j\in\mathcal{J}} O_{js} z_{jks} \\ &+ \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} f_{hkms} \quad \forall k \in \mathcal{H}, s \in \mathcal{S}\end{aligned} \tag{11}$$

$$q_k + \sum_{j\in\mathcal{J}} O_{js} z_{jks} + \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} f_{hkms} \leq \kappa_k \cdot (x_k + y_{ks}) \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{12}$$

$$x_k, y_{ks}, z_{iks}, z_{jks} \in \{0, 1\} \quad \forall i, j, k, s \tag{13}$$

$$q_k, f_{khms}, W_s \geq 0 \quad \forall k, h, m, s \tag{14}$$

Constraints (4)–(6) govern hub establishment and reactive-hub safe-zone placement, while per-scenario availability v_{ks} filters planned hubs in (8); (7)–(9) enforce single-allocation via available arcs; (10)–(12) maintain flow feasibility and capacity; (13)–(14) define variable domains.

3 Proposed Algorithm: PB-NSGA

The MO-IHLNDP is NP-hard, as its single-objective, single-scenario relaxation subsumes the uncapacitated facility location problem [18]. We therefore propose **PB-NSGA** (Priority-Based Nondominated Sorting Genetic Algorithm), which embeds the NSGA-II framework [14] with a custom heuristic decoder. The decoder is algorithm-agnostic: the same reconstruction and evaluation core is also embedded in GWO-HD and VNS-TS, so cross-algorithm differences later reflect search behavior rather than different recourse logic. We embed NSGA-II [14] rather than MOEA/D [19]: MOEA/D requires pre-specified decomposition weight vectors, presupposing the Pareto front geometry a priori; NSGA-II's crowding-distance selection preserves spread across the full Z_1–Z_2 spectrum without weight tuning; and the encoder-decoder design is algorithm-agnostic. The *decoder* is the contribution, and NSGA-II is the dominant operator in the humanitarian logistics MOEA literature [10, 15], enabling direct comparison.

3.1 Chromosome Encoding Scheme

Each chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$ consists of four segments. The binary vector $\mathbf{X} \in \{0, 1\}^{|\mathcal{H}|}$ determines planned hub establishment. The continuous vector $\mathbf{R} \in [0, 1]^{|\mathcal{H}|}$ defines the inventory fill fraction, setting $q_k = R_k \cdot \kappa_k$.

The integer vector $\mathbf{A} \in \{0, \ldots, |\mathcal{H}| - 1\}^{|\mathcal{I}|}$ encodes the preferred anchor hub for each demand i. The continuous vector $\mathbf{W} \in [0, 1]^6$ contains six heuristic weights governing the decoder: w_0 and w_3 dictate demand sorting priority by urgency $(\lambda \cdot D)$ and isolation; w_1, w_2, w_4 score candidate hubs by travel time, residual inventory, and preference for planned over reactive hubs; w_5 controls conservatism towards opening new reactive hubs.

3.2 Priority-Based Decoding Heuristic

The decoder is invoked once per chromosome per scenario $s \in \mathcal{S}$, reconstructing all second-stage decisions from $\mathbf{C}$ in four steps.

Step 1 – Initialisation. Planned hubs with $X_k = 1$ and $r_{ks} \leq \chi$ are activated with pre-positioned inventory $q_k = R_k \cdot \kappa_k$. If none qualify, the safest hub is forced active.

Step 2 – Demand Prioritisation. Each demand node i receives a composite score prioritising urgency, isolation, and proximity to active hubs. Demands are then processed in descending score order.

Step 3 – Tiered Hub Assignment. Each demand i is assigned to the best reachable active hub, scored by travel time and residual capacity, starting from its anchor π_{A_i}. If all candidates are at capacity (*Pass 2*), the assignment still proceeds but logs a Capacity Violation (CV) penalty. Only when no reachable active hub exists does the decoder open a new reactive hub as a last resort, incurring its setup cost and incrementing CV.

Step 4 – Supply Routing and Objective Accumulation. Origins supply the most under-supplied active hubs, and surplus is redistributed via discounted inter-hub trans-shipment (α). Expected objectives Z_1 and Z_2 are accumulated across all $s \in \mathcal{S}$. Algorithm 1 formalises these procedures.

Algorithm 1 Priority-Based Decoder (single scenario s)

$\mathcal{H}_{\text{act}} \leftarrow \{k : X_k = 1, r_{ks} \leq \chi\}$; $q_k \leftarrow R_k \kappa_k$; if empty, force-activate $\arg\min_k r_{ks}$
Score demands by urgency, isolation, proximity; sort descending
for each demand i (priority order) **do**
    $K \leftarrow \max(1, \lceil w_5 |\mathcal{H}| \rceil)$; trial $\leftarrow \pi_{A_i}$; *Pass 1*: $k^* \leftarrow \arg\max \text{HubScore}(k)$ over first K active reachable hubs with residual > 0
    if Pass 1 succeeds **then** assign $i \to k^*$; deduct residual
    else if any active reachable hub **then** *Pass 2*: assign to first such hub (CV permitted)
    else *Reactive*: open nearest safe inactive hub; add F^a_{ks}; $\text{CV}^s \mathrel{+}= 1$
    end if
end for
Route origin-to-hub supply and redistribute surplus via MCF on the hub sub-network; accumulate Z_1^s, Z_2^s
return $(Z_1^s, Z_2^s, \text{CV}^s)$

Steps 1–3 enforce single-allocation and arc-availability feasibility (Constraints (7)–(9)); with these decisions fixed, Step 4 reduces flow-balance (11) to a **minimum cost flow (MCF)** sub-problem (origins as sources, under-supplied hubs as sinks, lateral transshipment arcs as redistribution edges) solved exactly per scenario. The decoder therefore traverses the feasible region of the MILP and minimises Z_1 to near-optimality for any decoded hub configuration, a key driver of PB-NSGA's solution quality.

3.3 Genetic Operators

The initial population seeds $\mathbf{X}$ with 1 to $\lfloor 0.6|\mathcal{H}|\rfloor$ open hubs, $\mathbf{R}$ from discrete pre-positioning levels, and $\mathbf{A}$, $\mathbf{W}$ uniformly. Crossover (probability p_c) applies Uniform Crossover to $\mathbf{X}$ and $\mathbf{A}$, and SBX (index η_{SBX}) to $\mathbf{R}$ and $\mathbf{W}$. Mutation (base rate p_m, decaying linearly from p_m^{high} to p_m^{low} over G generations) applies bit-flip, random-reset, and polynomial perturbation to the respective segments. Survival follows NSGA-II elitist constrained dominance [14]: feasibility first, then non-dominated rank, then crowding distance with Hamming tiebreaking in $\mathbf{X}$-space. Rank-1 stagnation triggers increased tournament size and $\mathbf{W}$-mutation, following the adaptive elitist strategy of [20].

4 Computational Experiments

4.1 Experimental Setup and Datasets

Algorithms were implemented in C++17 and Python 3 (Intel i5-1135G7, 16 GB RAM), evaluated across 20 independent runs reporting mean $\pm$ std. Problem constants: $\chi = 0.7$, $\alpha = 0.6$, $\gamma = 3.0$ kg/person, $\lambda_0 = 0.8$. PB-NSGA parameters, tuned via grid search: $N = 200$, $G \in \{300, 500\}$ for CV-Small/Large, $p_c = 0.98$, linear mutation decay $p_m \in [0.40, 0.10]$, $\eta_{\text{SBX}} = 1.5$, $\eta_{\text{poly}} = 8$. Performance metrics are **Hypervolume (HV)** [21] and $\mathbf{IGD^+}$ [22], computed against the combined non-dominated reference front pooled across all algorithms. We construct two CV instances over four flood-prone provinces (Da Nang, Quang Nam, Thua Thien-Hue, Quang Ngai): **CV-Small** ($|\mathcal{I}| = 20$, $|\mathcal{H}| = 5$, $|\mathcal{J}| = 2$, $|\mathcal{S}| = 3$, district resolution) and **CV-Large** ($|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$, $|\mathcal{J}| = 12$, $|\mathcal{S}| = 3$, commune resolution). Node coordinates map to real administrative centroids and confirmed logistics facilities; scenario risks and road disruptions are calibrated via a multi-criteria flood vulnerability score [23]; three transport modes (truck, motorboat, helicopter) are modelled, with helicopter links capped at 15% of active arcs per scenario. Scenarios are drawn via SAA [24] from a profile bank spanning mild/severe/extreme flood regimes; CV-Large first-stage decisions $(\hat{x}, \hat{q})$ are further validated on an independent out-of-sample (OOS) set of $N' = 10$ adversarial extreme scenarios disjoint from training. Because this OOS holdout is deliberately extreme-only, we treat feasibility as the primary robustness metric and interpret large OOS Z_2 increases as stress amplification rather than prediction error.

Scenario-count sensitivity. Figure 1 reports a scenario-count sensitivity study ($K = 10$ replications, exact enumeration on CV-Small). The std of Z_2 contracts from $\pm$110k ($N = 3$) to $\pm$30k ($N = 30$) and HV std from 0.25 to 0.07, confirming SAA variance reduction; a transient peak at $N = 5$ reflects unlucky extreme-heavy draws. Our calibrated three-profile design achieves equivalent stability at one-third the scenario count.

4.2 Computational Results

We first evaluate PB-NSGA against four baselines on **CV-Small**: a **Greedy Heuristic** (a council of cost-safety managers using nearest-neighbour assignment), **VNS-TS** [9] (Variable Neighbourhood + Tabu Search from the flood-relief literature), **GWO-HD** [10] (Grey Wolf Optimizer with Hamming-distance moves), and the **MILP** solver [25] applying Adaptive Weighted Sum [26] to produce the reference Pareto front.

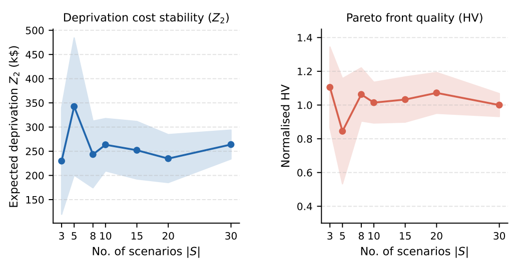

Fig. 1: Scenario-count sensitivity on CV-Small ($K = 10$ replications, exact enumeration). Std of Z_2 and HV decrease with N; the calibrated three-profile design achieves stability equivalent to $N{\approx}10$ random draws.

Table 1: Baseline comparison on CV-Small. HV and IGD$^+$ are normalized against the combined non-dominated reference front. Best values in **bold**.

Algorithm	HV $\uparrow$	IGD$^+$ $\downarrow$	CPU (s) $\downarrow$
Greedy	0.292	0.333	$<$ **0.1**
VNS-TS	0.777	0.117	60.0
GWO-HD	0.780	0.152	0.6
MILP (AWS)	**0.997**	**0.005**	4.9
PB-NSGA	0.925	0.039	1.1

PB-NSGA closes the gap to the MILP reference front (HV 0.925 vs. 0.997) in 1.1 s, a 4$\times$ runtime advantage over MILP and 50$\times$ over VNS-TS at competitive quality, demonstrating that the priority-based decoder effectively navigates the combinatorial structure of MO-IHLNDP.

On CV-Large, the same three search algorithms run on the full-scale instance ($|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$), where exact solvers become intractable; all other parameters follow Sec. 4.1.

Figure 2 shows Pareto fronts for both instances. On CV-Small, PB-NSGA closely tracks the MILP reference; GWO-HD converges to a similar region with more scatter; VNS-TS covers only the cost-efficient tail. On CV-Large (power

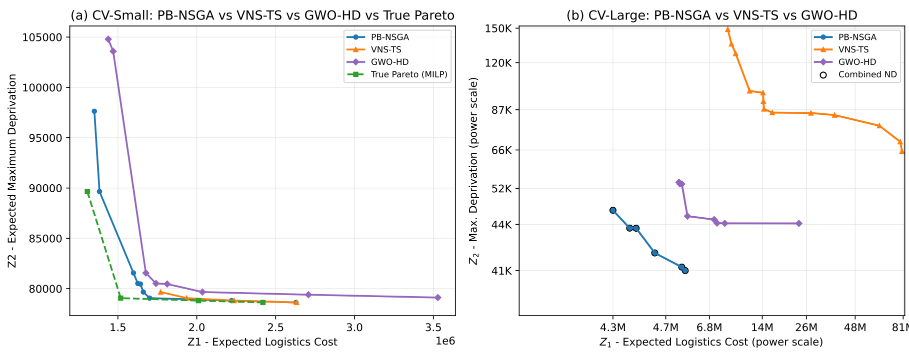

Fig. 2: Combined Pareto fronts. **Left:** CV-Small vs. MILP reference. **Right:** CV-Large (power scale), 20 matched runs per algorithm.

scale), PB-NSGA dominates the low-cost, low-deprivation region (Z_1: 4.3M–5M, Z_2: 41K–45K); GWO-HD extends the front rightward at higher Z_1 (up to 26M); VNS-TS is confined to the high-cost tail (Z_1 = 81M, Z_2 = 150K). Across 20 matched runs, PB-NSGA achieves $\mathrm{HV} = 1.000 \pm 0.006$ and $\mathrm{IGD}^+ = 0.000 \pm 0.001$ in 8.7 s, against 0.982 ± 0.007 in 4.8 s for GWO-HD and 0.823 ± 0.038 in 180.6 s for VNS-TS, a 20.7× runtime advantage at the best frontier quality.

Out-of-sample robustness vs. stratification. Operational viability under catastrophic stress complements objective-space convergence. Because all three algorithms share the same decoder, the 100% SAA-100 feasibility rate across all 20 seeds is attributable to the recourse engine, not the search. On the adversarial OOS set (10 extreme scenarios), PB-NSGA alone maintains perfect feasibility across all seeds; GWO-HD and VNS-TS remain highly feasible at 96.4% (95% CI [93.4%, 98.8%]) and 97.8% (95% CI [96.2%, 99.2%]), respectively. The feasibility gap reflects search quality, not recourse privilege.

Structural insights vs. reactive costs. Figure 3 visualizes the knee-point solution (Z_1 = \$4.6M, Z_2 = 43,453): eight planned hubs remain active with identical fill levels across all scenarios and no reactive hub is activated. At the CV-Large scale, dense pre-positioning at eight commune-resolution locations mathematically dominates the \$51K–\$109K per-scenario reactive setup cost, demonstrating economic rationality rather than a decoder bias. The recourse mechanism (Algorithm 1, Pass 3) remains available and verifiably triggers on sparser CV-Small instances under severe and extreme scenarios. Adaptation across severity regimes is modal rather than structural: helicopter routes displace truck links in the extreme scenario.

H16 (91%) and H15 (89%) carry the heaviest pre-positioned loads, anchoring coverage for the central and southern coastal corridor. The primary adaptation across scenarios is modal: mild and severe conditions are served by a mix of truck and watercraft links, while the extreme scenario activates helicopter routes (orange dashed) to reach high-risk demand nodes that become inaccessible by road and water.

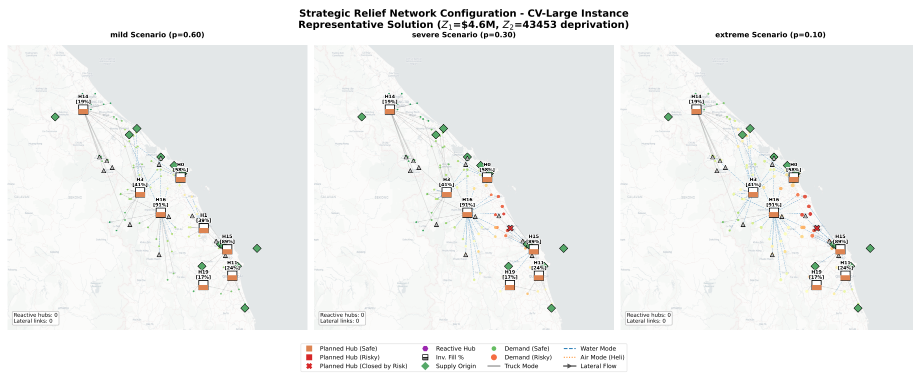

Fig. 3: CV-Large knee-point relief network across the three flood scenarios. Brackets: hub fill levels. Line styles: solid gray (truck), blue dashed (water), orange dashed (helicopter).

Comparing the calibrated $N = 3$ design against a tier-balanced $N = 10$ set across 20 PB-NSGA runs each, $N = 3$ achieves superior OOS feasibility (1.000 vs. 0.931; $\Delta = -0.069$, 95% CI $[-0.128, -0.029]$) at one-third the computational burden. The $N = 10$ design occasionally yields lower typical degradation, but introduces severe robustness variance evidenced by a non-significant mean OOS ΔZ_2 driven by catastrophic outliers (up to 22,037% error). Feasibility therefore remains our primary robustness metric, validating deliberate stratification over uniform random sampling.

Decision support. Hub selection frequencies across 128 Pareto solutions reveal a rigid investment hierarchy. Coastal nodes with multi-modal accessibility (H15, H16) form an uncompromisable core (100% activation). Inland depots (H1, H3, H9) act as conditional assets whose utility depends strictly on the risk threshold χ: H1's risk escalates from 0.46 (mild) to 0.94 (severe), and the decoder bypasses it whenever $r_{ks} > \chi$ (Pass 1), motivating contingency pre-staging at H14 (risk ≤ 0.13 across all regimes). Joint Z_1–Z_2 optimisation delivers a 3.5× equity improvement (worst-case deprivation 43k vs. 150k for the cost-only Pareto extreme) at $< 25\%$ logistics-cost premium.

5 Conclusion and Future Work

For Central Vietnam, the framework converts the abstract incomplete-network formulation into actionable infrastructure guidance. Regional resilience is best achieved by anchoring operations at coastal, multi-modal nodes capable of bypassing severed road networks; inland depots represent conditional assets whose utility degrades sharply under severe risk thresholds, requiring planners to shift contingency inventory to lower-risk, secondary inland staging areas. The dominance of pre-positioning over reactive activation at this density indicates that, once a network exceeds a critical hub-to-demand ratio, the marginal value of reactive setup investments falls below their cost; reactive-hub capital should be redirected toward pre-positioning depth and modal diversification (marine and rotary-wing access at coastal anchors). Algorithmically, PB-NSGA's matched 20-seed advantage is feasibility-driven, not cost-driven: on the adversarial OOS set its 100% feasibility rate distinguishes it from GWO-HD and VNS-TS, which share the same priority-based decoder but differ in their search operators. Since recourse logic is held constant, the feasibility gap isolates search efficiency as the operative factor. The principal limit is the variance of Z_2 under uniform-probability training; calibrated stratification removes it, but operational deployments with poorly characterised flood frequencies will inherit this variance until field data refine the profile distribution.

Limitations and future work. The model uses a synthetic instance calibrated from administrative and geographic data; field validation against actual disaster records is needed before deployment. Three directions are prioritised: integrating real-time sensor and satellite feeds; penalising hub-processing degradation under high-risk conditions to bridge scenario planning and tail-risk management; and scaling PB-NSGA to hour-long runtime budgets for regional deployments.

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